FleetLine E1 — Road to 2035

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PHASE 1  2026-27          GATE              PHASE 2  2028-30       GATE        PHASE 3
 2031-35
  Delhi NCR                 G1 CELL           Delhi                    G4 SCALE    Pan-
 India
  1,000-2,000 veh           G2 DUTY           Bengaluru                G5 CELLS    100,
 000+ cum.
  margin segment            G3 BATTERY        Mumbai
    |                         |                 |                          |
 |
    └── 5 gates, all measurements — not dates ──────────────────────────────┘
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Gates are **measurements, not calendar milestones.** Each has a threshold and a failure action.

Gate	Measure	Threshold	If it fails
G1 CELL	Qualified LFP sources	Two, price parity within 10%	Stay dual-imported; do not commit Phase 2 volume
G2 DUTY	Measured paid km/day	≥ 200	Re-run the TCO before scaling — the case is duty-sensitive
G3 BATTERY	Per-cell SoH telemetry	12 months, closing the **3.6×** disagreement in published LFP time-to-80%	Capacity-retention guarantee is unsignable
G4 SCALE	Cumulative fleet	20,000 vehicles, 21.9 h/month downtime slack held against a 25–40% public-charger session-failure rate	Hold at Phase 2 footprint
G5 CELLS	Domestic LFP capacity	**>20 GWh operating, automotive-qualified**	Localisation claim stays at pack assembly

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Phase 1 · 2026–27 · Delhi NCR

- **1,000–2,000 vehicles** — indicative and gate-conditional, not a forecast
- **Segment:** airport transfers + corporate commute. Launch here because EV adoption is 18% against 10% street-hail, and dispatch is pre-booked
- **Charging:** 1 × 7.4 kW AC port per vehicle, 10% N+1 spares, **₹99,000/vehicle installed**

  Overnight window 25.68 kWh ÷ (7.4 kW × 0.90) = **3.86 h**. One port is enough. No DC fast-charging required
- **Cells:** Reliance Jamnagar LFP primary, qualified import as bridge, dual-sourced from day one

- **Power:** ₹4.50/kWh Delhi BEE EV tariff

Phase 2 · 2028–30 · Delhi + Bengaluru + Mumbai

All three cities clear at five years on the published basis. Cumulative 15,000–25,000 vehicles.

Phase 3 · 2031–35 · Pan-India

100,000+ cumulative. Deep localisation proceeds **only** if G5 is met.

Localisation

	Window	Content
L1	2026–27	Pack assembly, BMS, thermal management, vehicle integration
L2	2028–30	Domestic LFP cells for a share of volume; imported for the balance
L3	2031+ **if gated**	Domestic cells plus CAM, AAM, separator content

The honest ceiling. ACC PLI has delivered **1.4 GWh of 50 GWh — 2.8%** — and all of it from one company. Domestic value addition sits below the 60% DVA threshold because CAM, AAM, separators, electrolytes and foils are imported. Reliance Jamnagar is the LFP player (40 GWh, ramping through H2 2026). Ola Cell is the only Indian producer at scale and is NMC-first — the wrong chemistry for this vehicle, which is why dual-sourcing is the plan and not the exception.

Infrastructure

- **One AC port per vehicle.** CCS2 mandated for four-wheelers by the **2022 amendment to AIS-138 Part 2**; OCPP 1.6J
- **Range envelope: 268 km at 15% reserve** (315 km at zero reserve)
  - Feasible one-way: Delhi–Agra 200 · Mumbai–Pune 150 · Mumbai–Nashik 170 · Bengaluru–Mysuru 145
  - Marginal: Delhi–Jaipur 280 — 88.8% of usable pack
  - **Beyond ~300 km, destination charging is mandatory — a fleet-contract requirement, not a vehicle spec**
- **25–40% public-charger session-failure rate** (300,000+ model/OEM/CPO combinations) against **21.9 h/month** of real downtime slack. This is the operational risk that binds, not range

Policy

State	What it gives	Expiry
Delhi	0% road tax for the life of the vehicle; ₹1 lakh scrapping incentive. **No purchase incentive for electric cars**	**31-Mar-2030**
Maharashtra	**15% of ex-factory, capped ₹2 lakh, transport e-4W — 25,000 units**; 100% toll exemption on the Mumbai–Pune Expressway and Samruddhi Mahamarg; EV road-tax exempt	**31-Mar-2030**
Karnataka	20–25% capex subsidy, stamp-duty reimbursement, three plug-and-play clusters. No consumer incentive	2030
Central	**PM E-DRIVE has no e-4W category at all** — ₹11,900 cr covers 2W, 3W, rickshaws, ambulances, trucks, buses	31-Mar-2028

Delhi and Maharashtra both expire five years before the 2035 target. Renewing them is a policy position, not an engineering one, and it is worth roughly 5 points on the five-year case.

Economics

NPV at a 12% hurdle, escalated, financed 60 months, **including the ₹99,000 charger**.

	Delhi	Mumbai	Bengaluru
5-year	**+6.7%**	**+4.0%**	**+0.9%**
8-year	**+14.6%**	**+10.6%**	**+9.0%**
2035, 5-year	+15.7%	+10.4%	+9.0%
2035, 8-year	+23.1%	+16.4%	+16.3%

One negative cell in the whole 48-cell model: Bengaluru, 5 years, cash purchase, −0.4%.
Where it breaks. Power-escalation break-even at five years is **Delhi 14.5%/yr, Mumbai 8.8%, Bengaluru 8.7%**. Electricity escalation is unambiguously bad for the EV in every city — worst in Mumbai at −9.2 points — and the CNG leg is what carries the case. Do not present escalation as good news.

Every figure regenerates from `MACH1_WORKING_FILES/tco_engine/` *(sections A–R). Detailed sourcing in* `MACH1_IMPLEMENTATION_ROADMAP_DETAILED_SUPERSEDED.md` *and* `MACH1_TCO_ANALYSIS.md`.